

Informe LAB O2 Haskell

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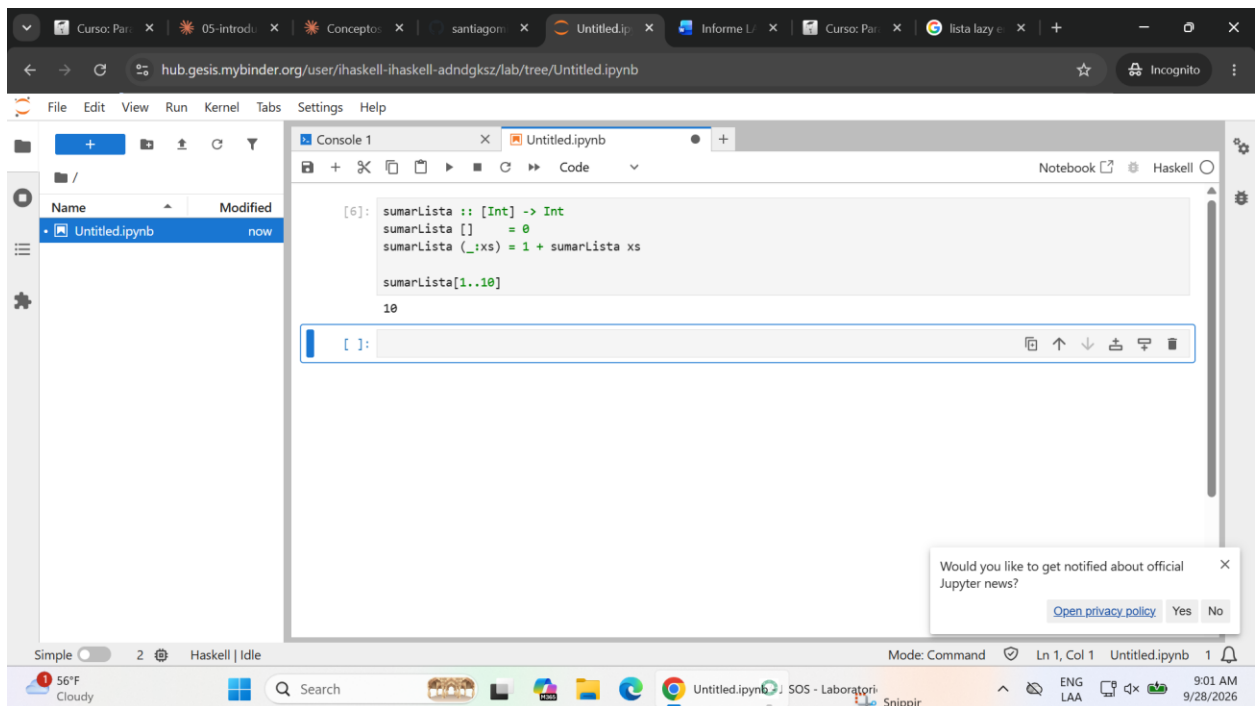
Paradigmas de la Programación

Juan Camilo Ángel Hernández

5 de octubre de 2026

Ejercicio 1 - Manejo de listas

- 1) Definir las siguientes funciones usando recursión.
 - a) Retorna la longitud de una lista



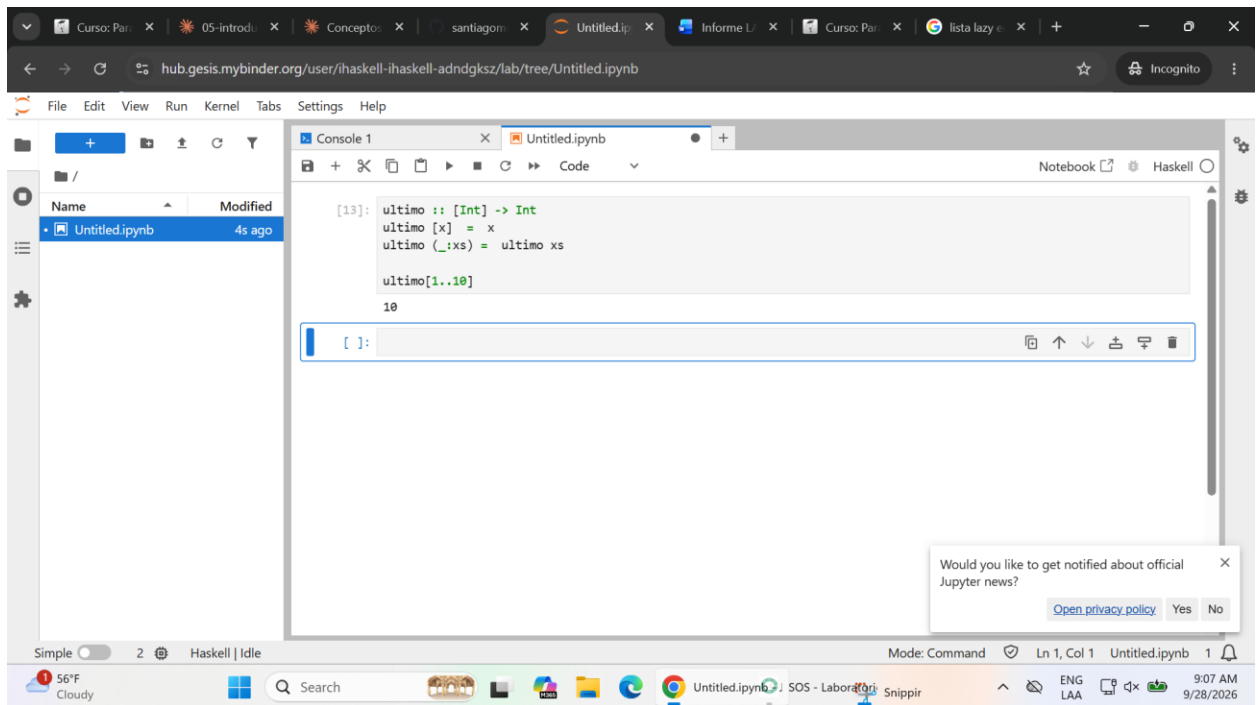
The screenshot shows a web browser window with a Jupyter Notebook interface. The browser's address bar displays the URL: `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgkszl/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with a single file named `Untitled.ipynb`. The main area of the notebook is divided into a code editor and a console. The code editor contains the following Haskell code:

```
[6]: sumarLista :: [Int] -> Int
    sumarLista [] = 0
    sumarLista (_:xs) = 1 + sumarLista xs

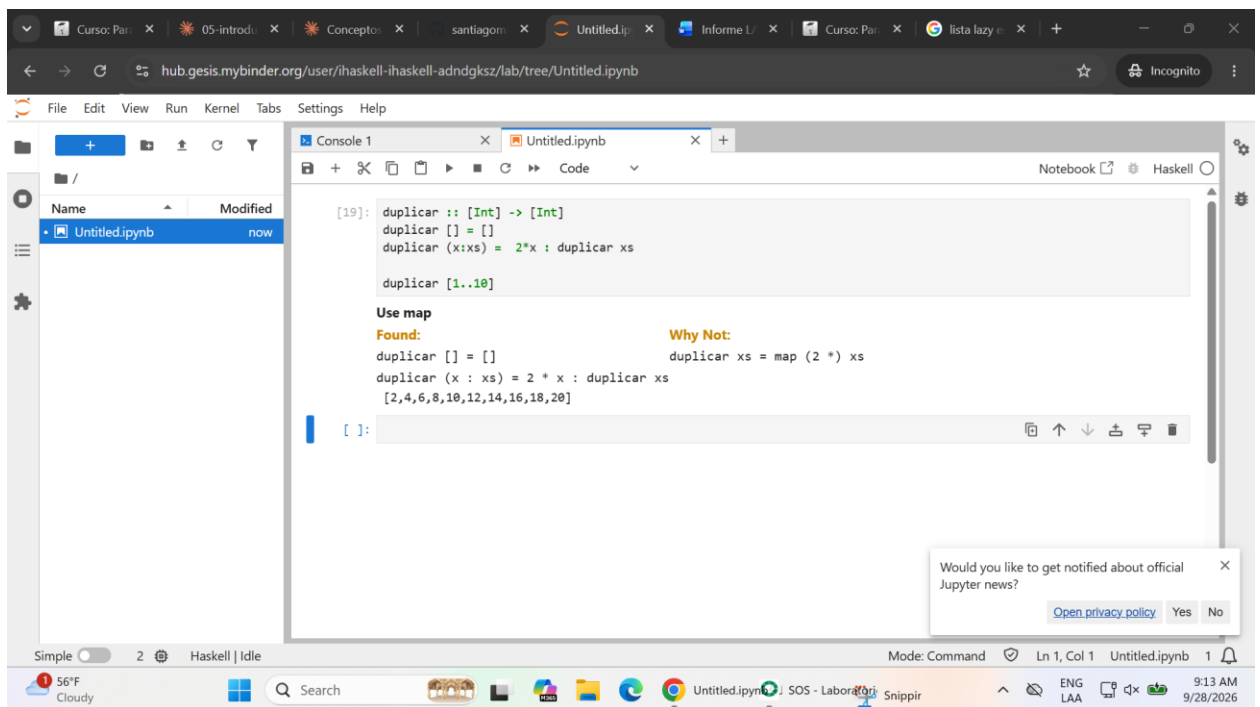
    sumarLista[1..10]
```

The console shows the output of the last line of code, which is `10`. A small notification box in the bottom right corner asks, "Would you like to get notified about official Jupyter news?" with buttons for "Open privacy policy", "Yes", and "No". The bottom status bar of the notebook indicates the mode is "Command", the cursor is at "Ln 1, Col 1", and the file is "Untitled.ipynb". The system tray at the bottom of the screen shows the date and time as "9:01 AM 9/28/2026".

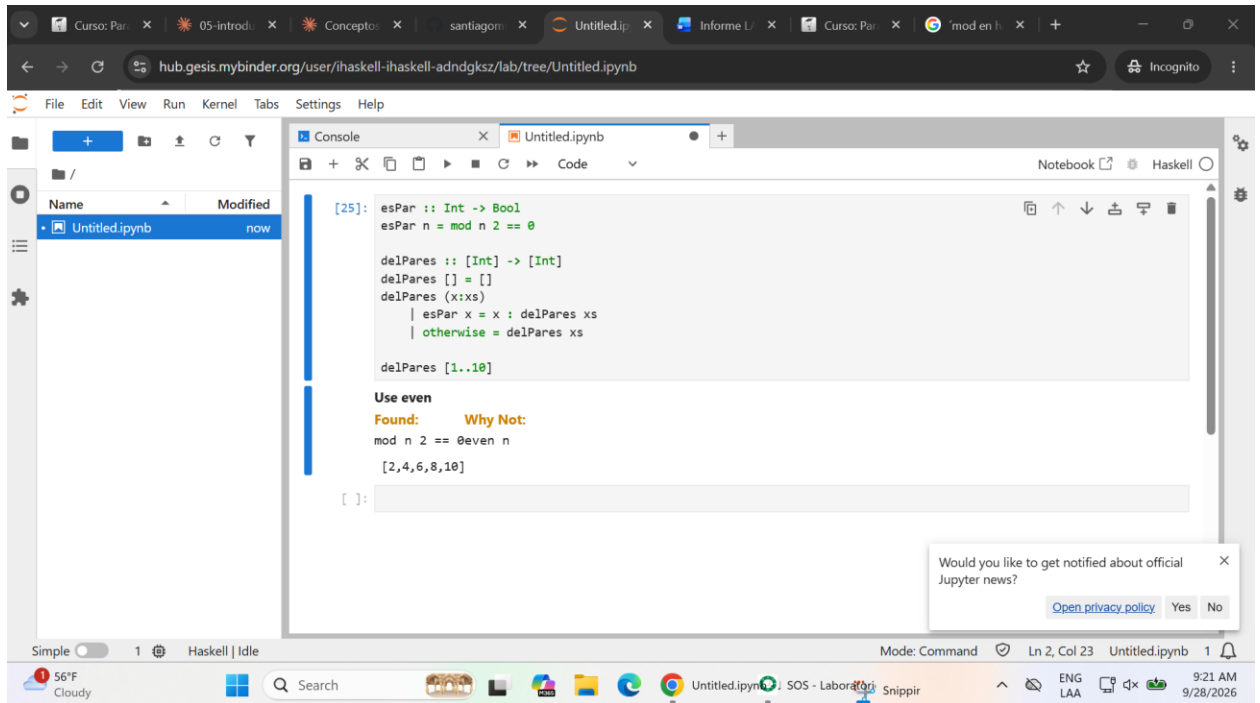
- b) Retorna el último elemento de una lista



c) Duplica cada elemento de una lista de números

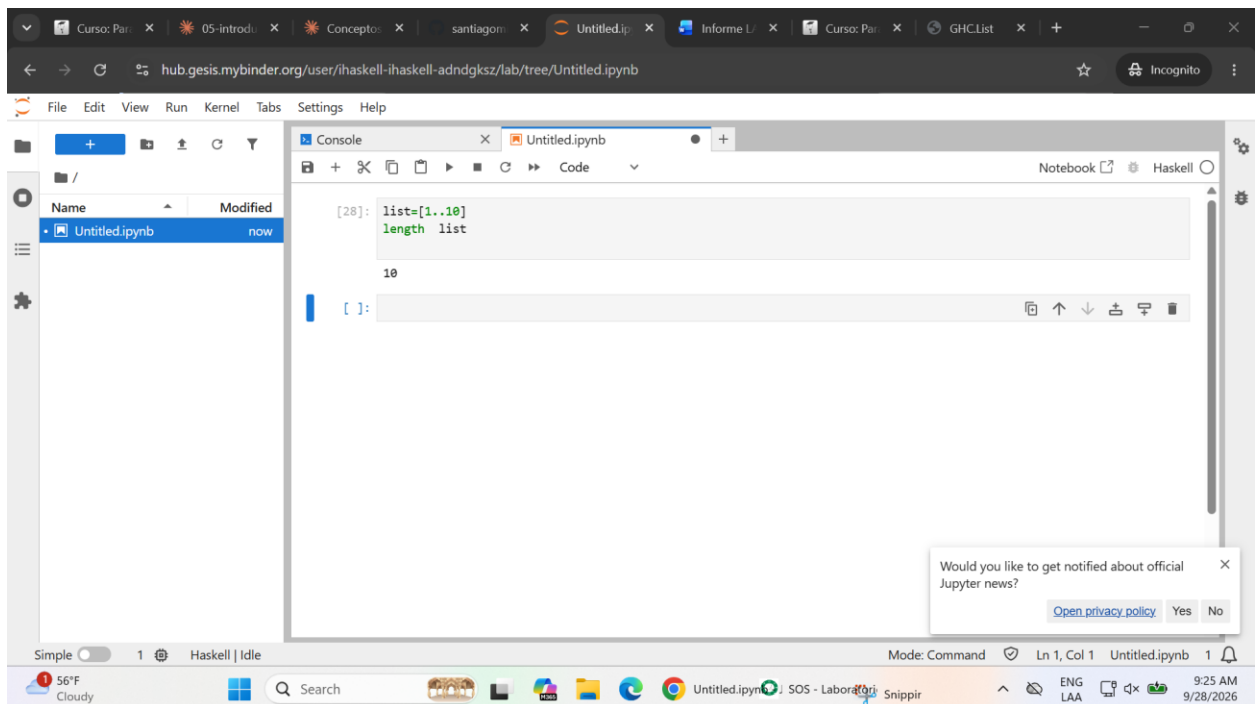


d) Eliminar los elementos pares de una lista

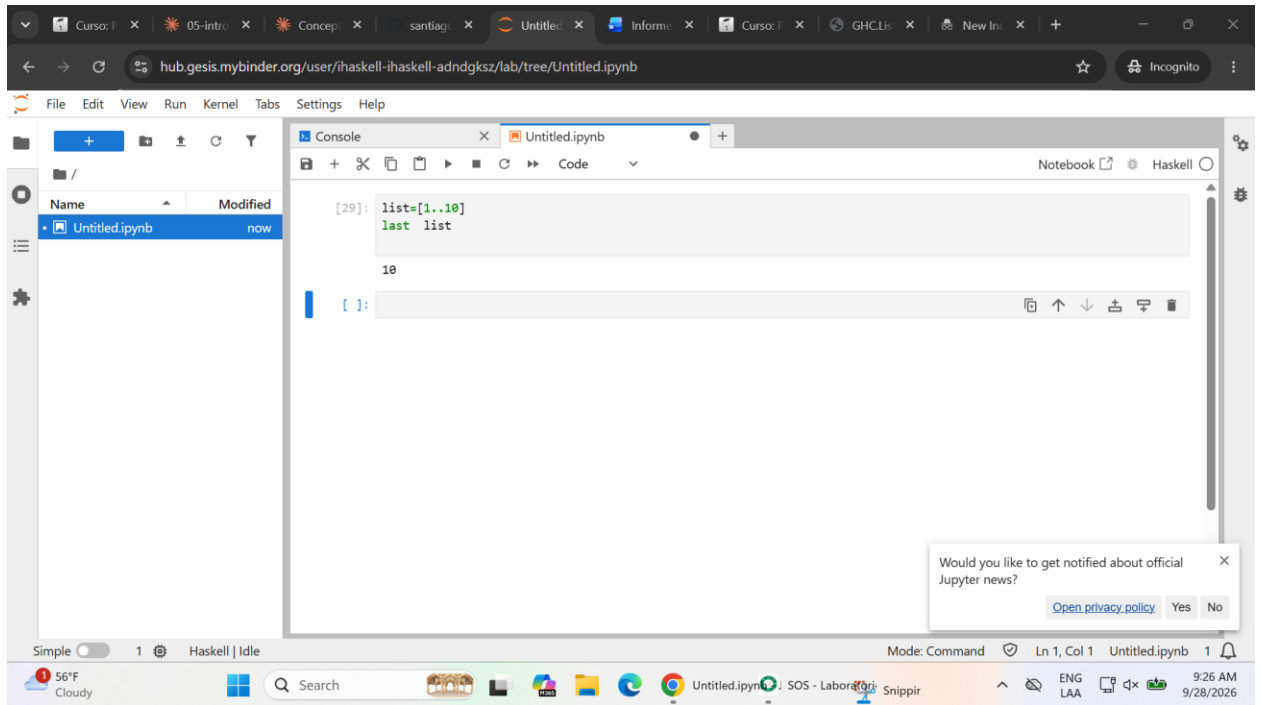


2) Define estas mismas funciones pero ahora usando funciones de orden superior como map y filter

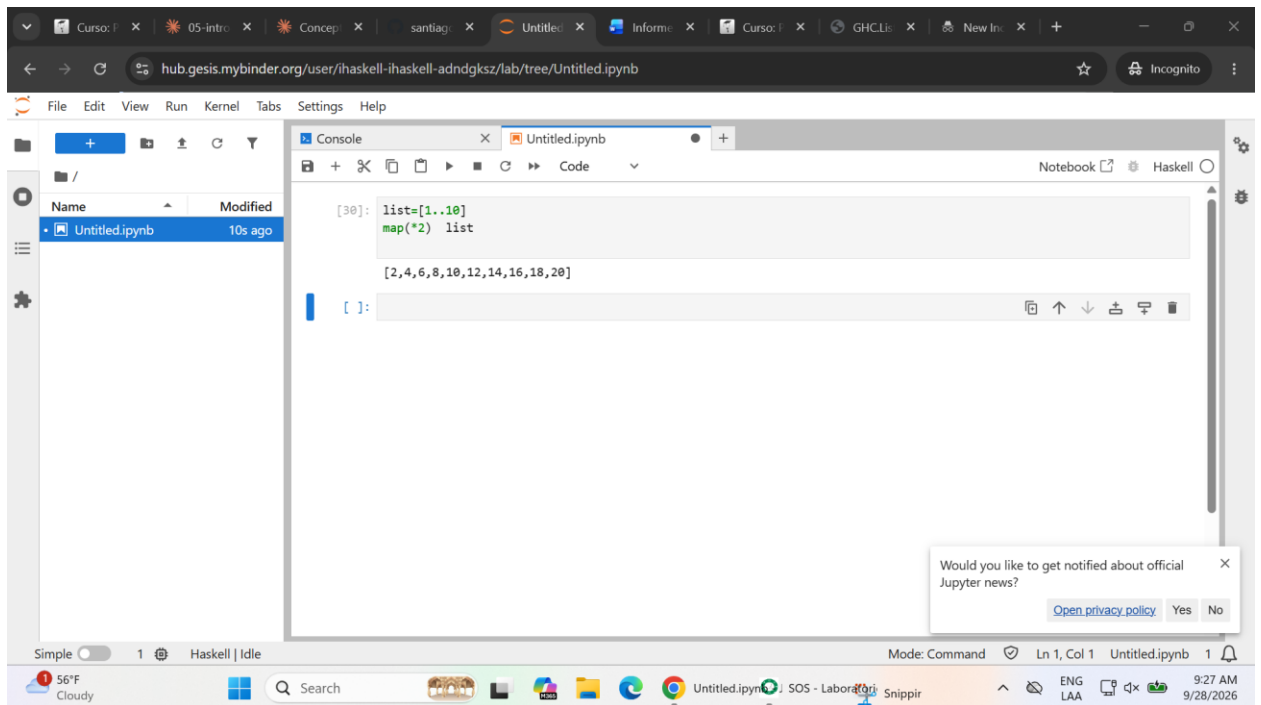
a) Retorna la longitud de una lista



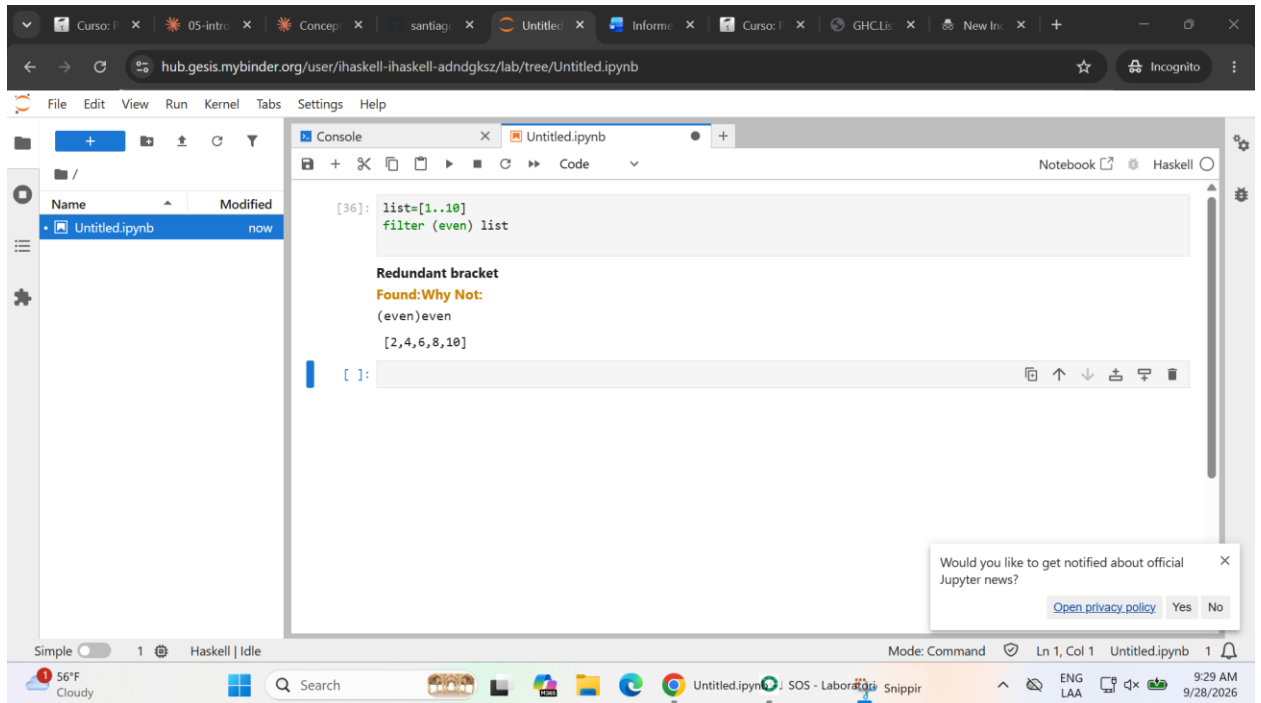
b) Retorna el último elemento de una lista



c) Duplica cada elemento de una lista de números

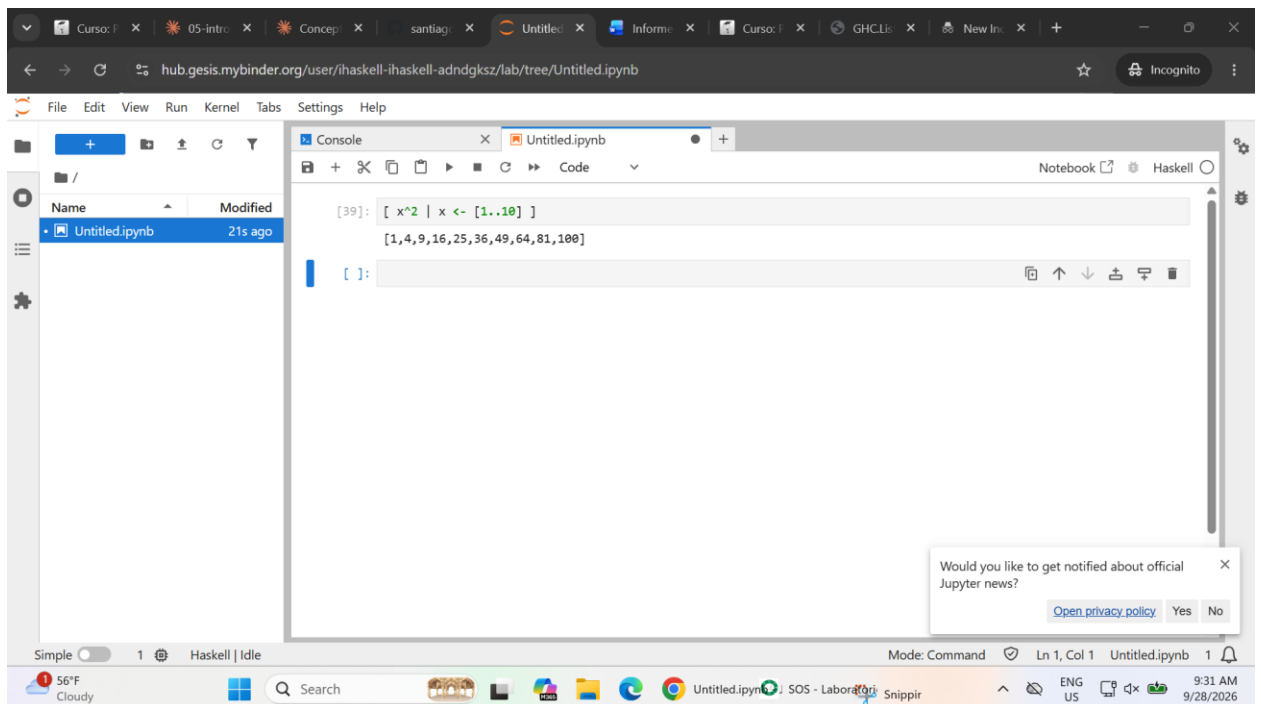


d) Eliminar los elementos pares de una lista



Ejercicio 2 - Listas por comprensión

- 1) Usa comprensión de listas para:
 - a) Obtener los cuadrados de los primeros 10 números.



b) Filtrar los divisibles por 3 entre 1 y 30.

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The code editor contains the following code:

```
[42]: [ x | x <- [1..30] , mod x 3 == 0 ]
```

The output of the code is displayed below the code cell:

```
[3,6,9,12,15,18,21,24,27,30]
```

A notification bubble in the bottom right corner asks: "Would you like to get notified about official Jupyter news?" with links for "Open privacy policy", "Yes", and "No".

c) Generar una lista de pares (x, y) donde $x < y$ y ambos estén en [1..5].

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The code editor contains the following code:

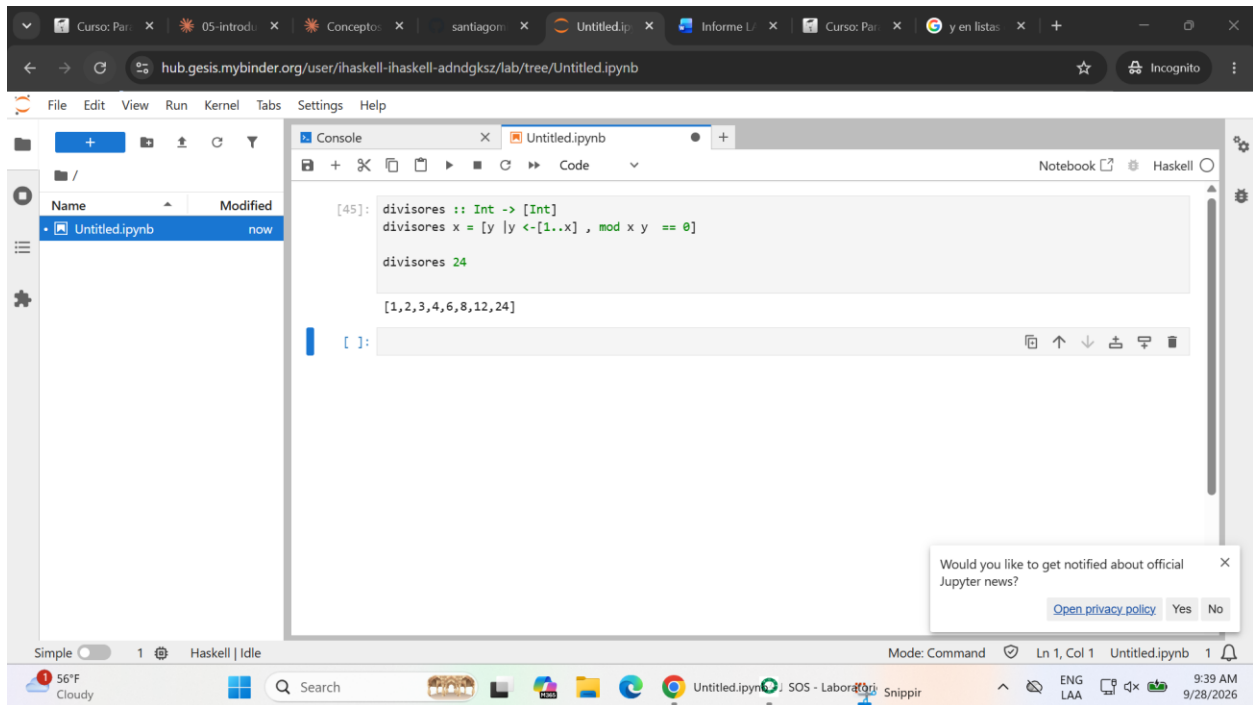
```
[43]: [ (x,y) | x <- [1..5] , y <- [1..5]
```

The output of the code is displayed below the code cell:

```
[(1,1),(1,2),(1,3),(1,4),(1,5),(2,1),(2,2),(2,3),(2,4),(2,5),(3,1),(3,2),(3,3),(3,4),(3,5),(4,1),(4,2),(4,3),(4,4),(4,5),(5,1),(5,2),(5,3),(5,4),(5,5)]
```

A notification bubble in the bottom right corner asks: "Would you like to get notified about official Jupyter news?" with links for "Open privacy policy", "Yes", and "No".

- 2) Define una función que devuelva los divisores de un número (usando lista por comprensión):



The screenshot shows a Jupyter Notebook interface in a web browser. The browser's address bar displays the URL: `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with a single file named `Untitled.ipynb`. The main area of the notebook contains a code cell with the following Haskell code:

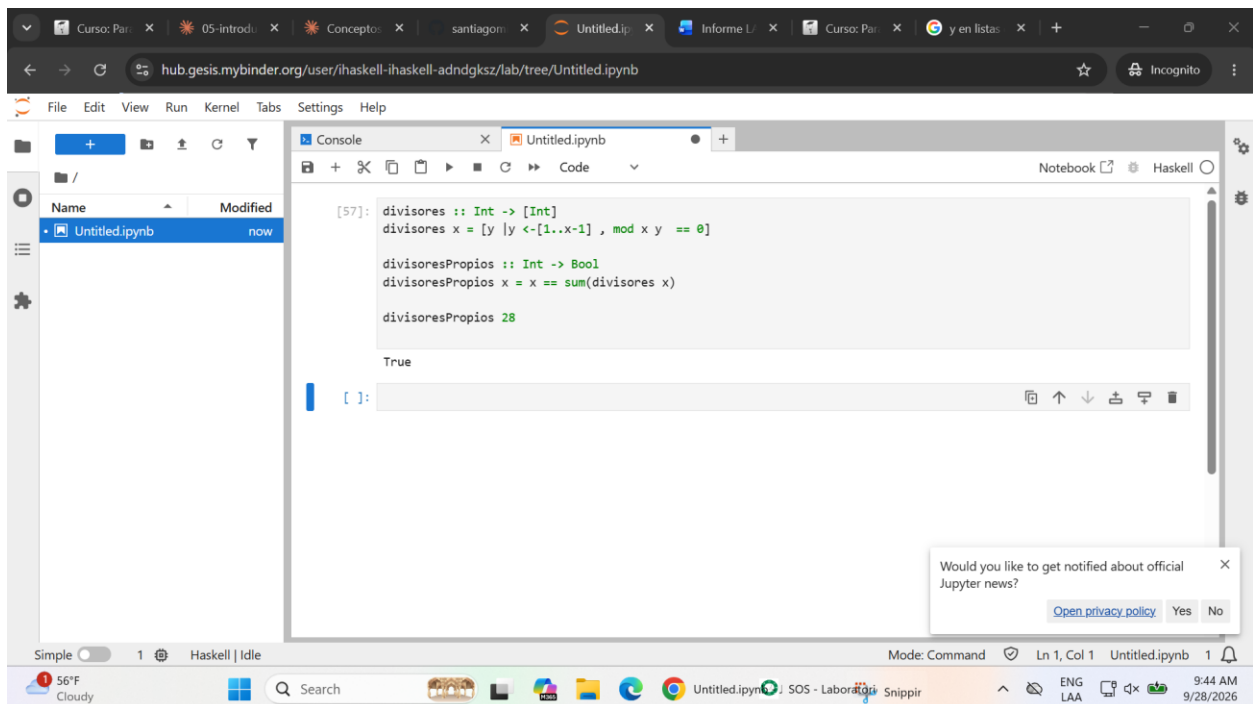
```
[45]: divisores :: Int -> [Int]
      divisores x = [y | y <- [1..x], mod x y == 0]

      divisores 24

      [1,2,3,4,6,8,12,24]
```

The output of the code cell is the list `[1,2,3,4,6,8,12,24]`. A notification bubble in the bottom right corner asks, "Would you like to get notified about official Jupyter news?" with links for "Open privacy policy", "Yes", and "No". The bottom status bar indicates the mode is "Command" and the kernel is "Haskell | Idle".

- 3) Define la función `divisoresPropios`, crea una función que determine si un número es perfecto (suma de sus divisores propios = número):



The screenshot shows the same Jupyter Notebook interface. The code cell now contains the following Haskell code:

```
[57]: divisores :: Int -> [Int]
      divisores x = [y | y <- [1..x-1], mod x y == 0]

      divisoresPropios :: Int -> Bool
      divisoresPropios x = x == sum(divisores x)

      divisoresPropios 28

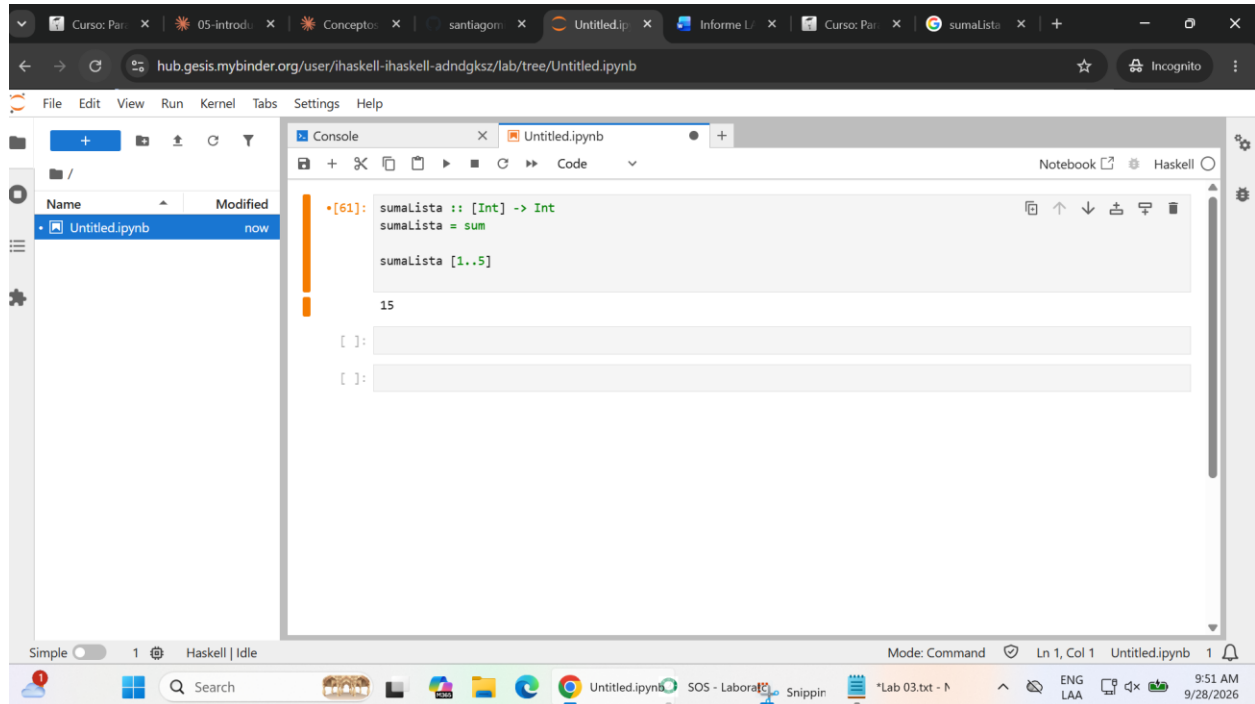
      True
```

The output of the code cell is the boolean value `True`. The notification bubble and status bar are identical to the previous screenshot.

Ejercicio 3 - Funciones de orden superior

1) Reescribe las siguientes funciones sin recursión explícita, usando funciones de orden superior:

a) Suma de todos los elementos

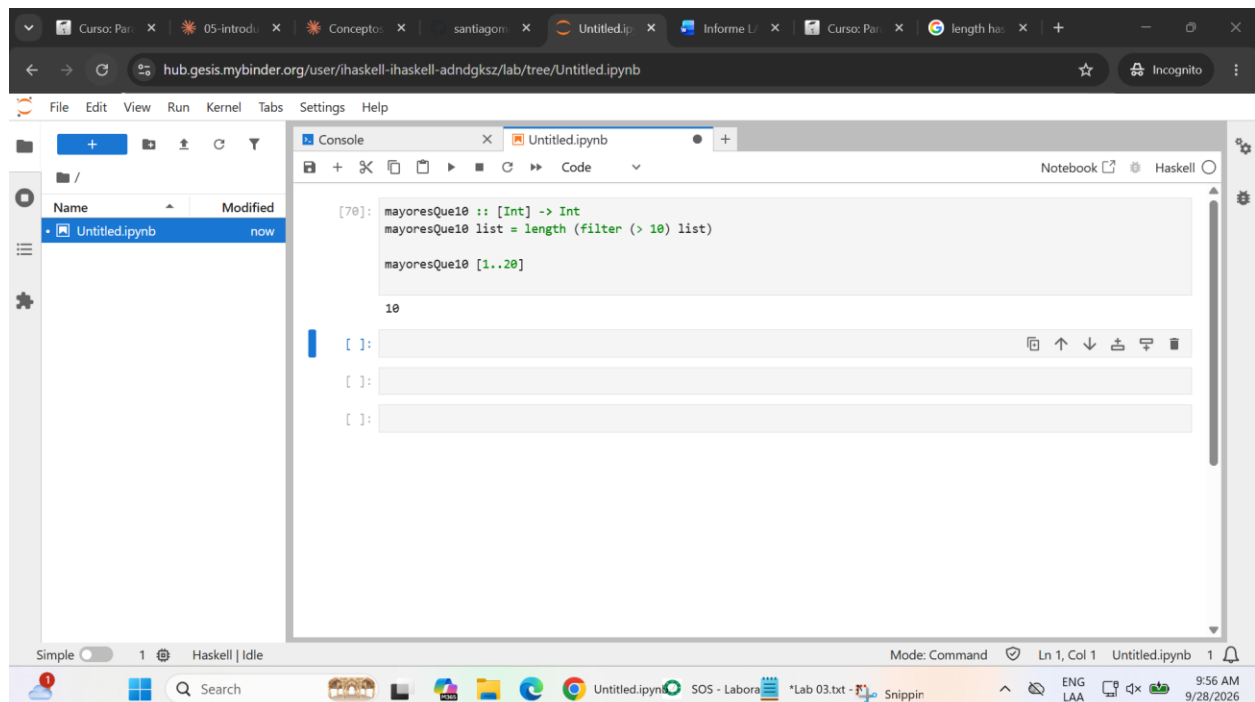


The screenshot shows a web browser window with a Jupyter Notebook interface. The browser's address bar displays the URL `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with `Untitled.ipynb` selected. The main editor area contains the following Haskell code in a code cell:

```
[61]: sumaLista :: [Int] -> Int
      sumaLista = sum
      sumaLista [1..5]
```

The output of the code cell is the integer `15`. The notebook's status bar at the bottom indicates the mode is 'Command' and the current position is 'Ln 1, Col 1'.

b) Contar cuántos elementos son mayores que 10

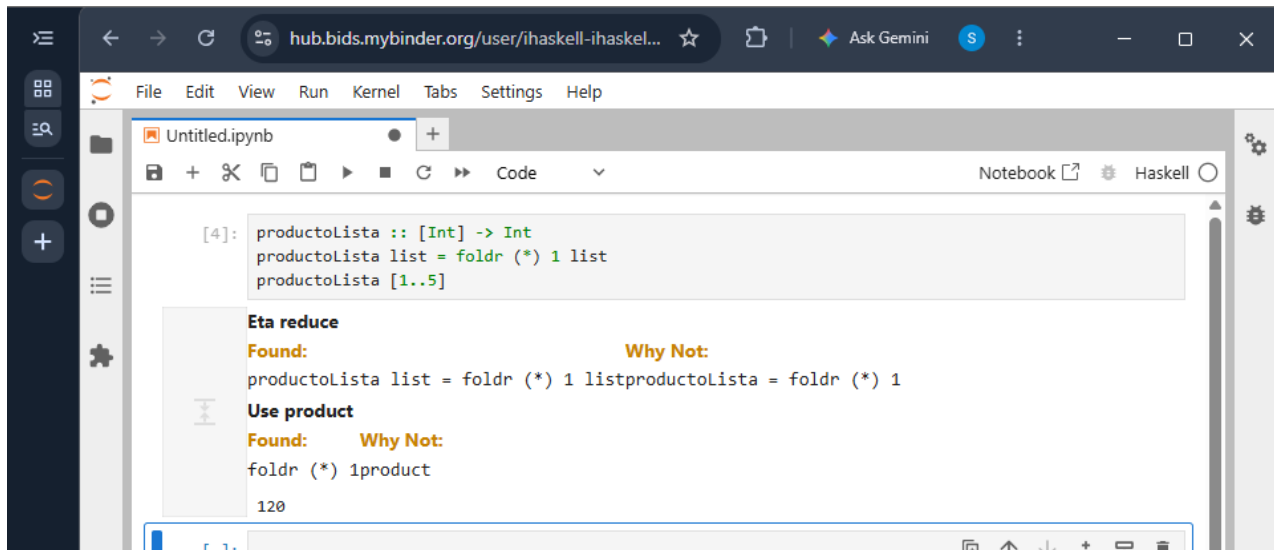


The screenshot shows a web browser window with a Jupyter Notebook interface. The browser's address bar displays the URL `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with `Untitled.ipynb` selected. The main editor area contains the following Haskell code in a code cell:

```
[70]: mayoresQue10 :: [Int] -> Int
      mayoresQue10 list = length (filter (> 10) list)
      mayoresQue10 [1..20]
```

The output of the code cell is the integer `10`. The notebook's status bar at the bottom indicates the mode is 'Command' and the current position is 'Ln 1, Col 1'.

c) Calcular el producto de una lista de números

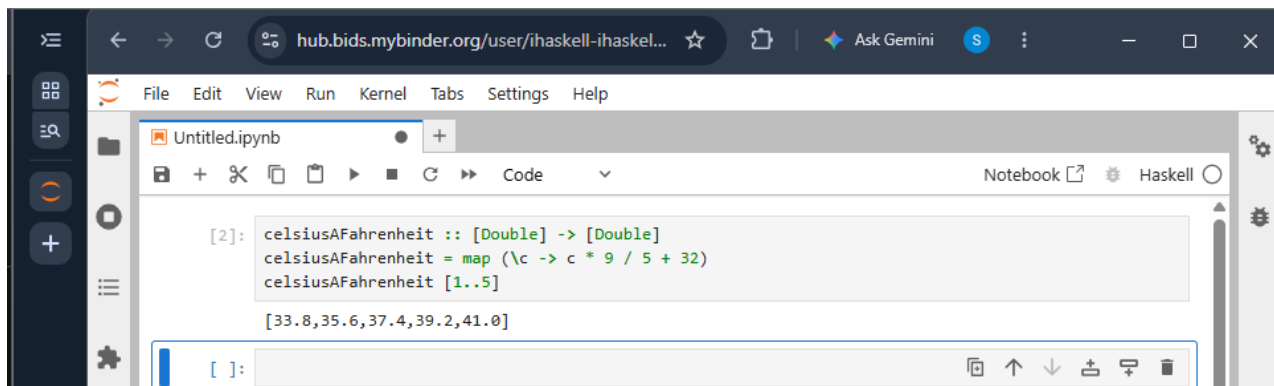


The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The notebook interface has a menu bar with File, Edit, View, Run, Kernel, Tabs, Settings, and Help. The notebook title is "Untitled.ipynb". The code cell contains the following Haskell code:

```
[4]: productoLista :: [Int] -> Int
     productoLista list = foldr (*) 1 list
     productoLista [1..5]
```

The output shows the result of the foldr function, which is 120. Below the code, there is a section titled "Eta reduce" with two columns: "Found:" and "Why Not:". The "Found:" column contains the code `productoLista list = foldr (*) 1 list` and the "Why Not:" column contains the code `productoLista = foldr (*) 1`. Below this, there is a section titled "Use product" with two columns: "Found:" and "Why Not:". The "Found:" column contains the code `foldr (*) 1 product` and the "Why Not:" column contains the code `120`.

d) Convierte una lista de grados Celsius a Fahrenheit

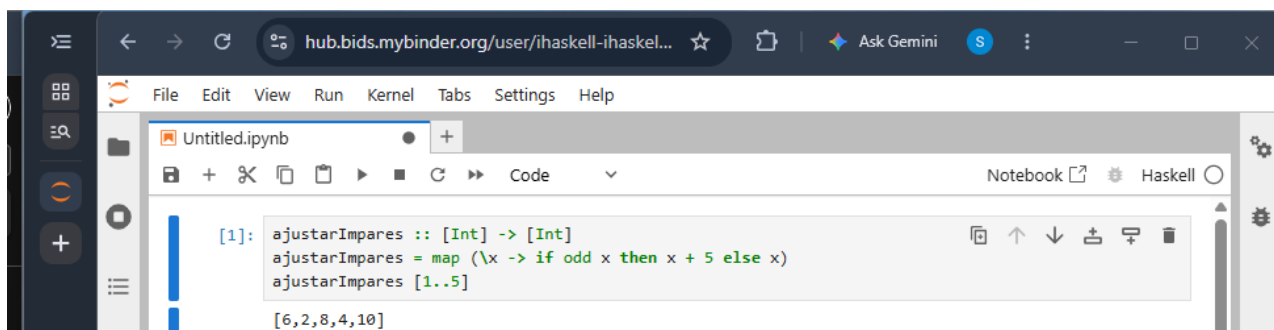


The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The notebook interface has a menu bar with File, Edit, View, Run, Kernel, Tabs, Settings, and Help. The notebook title is "Untitled.ipynb". The code cell contains the following Haskell code:

```
[2]: celsiusAFahrenheit :: [Double] -> [Double]
     celsiusAFahrenheit = map (\c -> c * 9 / 5 + 32)
     celsiusAFahrenheit [1..5]
```

The output shows the result of the function, which is `[33.8,35.6,37.4,39.2,41.0]`.

e) Suma 5 a cada número impar de una lista y deja igual los pares.



The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The notebook interface has a menu bar with File, Edit, View, Run, Kernel, Tabs, Settings, and Help. The notebook title is "Untitled.ipynb". The code cell contains the following Haskell code:

```
[1]: ajustarImpares :: [Int] -> [Int]
     ajustarImpares = map (\x -> if odd x then x + 5 else x)
     ajustarImpares [1..5]
```

The output shows the result of the function, which is `[6,2,8,4,10]`.

2) Define las siguientes funciones usando filter

a) Devuelve solo los nombres con más de 5 letras.

The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The browser's address bar also contains "Ask Gemini" and a blue "S" icon. The Jupyter Notebook interface has a menu bar with "File", "Edit", "View", "Run", "Kernel", "Tabs", "Settings", and "Help". The notebook title is "Untitled.ipynb". The code editor shows the following Haskell code:

```
[3]: nombresLargos :: [String] -> [String]
nombresLargos = filter (> 5) . length
nombresLargos ["Ana", "Alejandro", "Luis", "Fernanda", "Pedro"]
```

The output of the code is displayed below the code cell:

```
["Alejandro", "Fernanda"]
```

b) Filtra los números negativos de una lista.

The screenshot shows the same web browser window and Jupyter Notebook interface. The code editor shows the following Haskell code:

```
[6]: soloNegativos :: [Int] -> [Int]
soloNegativos = filter (< 0)
soloNegativos [-5..5]
```

The output of the code is displayed below the code cell:

```
[-5,-4,-3,-2,-1]
```

c) De una lista de edades, obtiene las que estén entre 18 y 25 años.

The screenshot shows the same web browser window and Jupyter Notebook interface. The code editor shows the following Haskell code:

```
[10]: edadesUniversitarios :: [Int] -> [Int]
edadesUniversitarios = filter (\e -> e >= 18 && e <= 25)
edadesUniversitarios [0..25]
```

The output of the code is displayed below the code cell:

```
[18,19,20,21,22,23,24,25]
```

3) Define las siguientes funciones usando filter y map

a) Dada una lista de precios, sumar IVA (19%) solo a los valores mayores o iguales a 1000.

The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The browser has tabs for "Ask Gemini" and a search bar. The Jupyter Notebook interface includes a menu bar (File, Edit, View, Run, Kernel, Tabs, Settings, Help) and a toolbar with icons for file operations and execution. The notebook is titled "Untitled.ipynb" and is in "Code" view. The code cell [13] contains the following Haskell code:

```
agregarIVA :: [Double] -> [Double]
agregarIVA = map (\e -> if e >= 1000 then e+(e*0.19) else e)
agregarIVA [1000,2300,300]
```

The output of the code is `[1190.0,2737.0,300.0]`.

b) De una lista de palabras, obtener solo las que comienzan con vocal y devolver su longitud.

The screenshot shows the same web browser window. The Jupyter Notebook is titled "Untitled.ipynb" and is in "Code" view. The code cell [15] contains the following Haskell code:

```
empiezaConVocal :: String -> Bool
empiezaConVocal [] = False
empiezaConVocal (c:_) = c `elem` "aeiouAEIOUáéíóúÁÉÍÓÚ"

longitudesVocales :: [String] -> [Int]
longitudesVocales = map length . filter empiezaConVocal
longitudesVocales ["hola","ola"]
```

The output of the code is `[3]`.

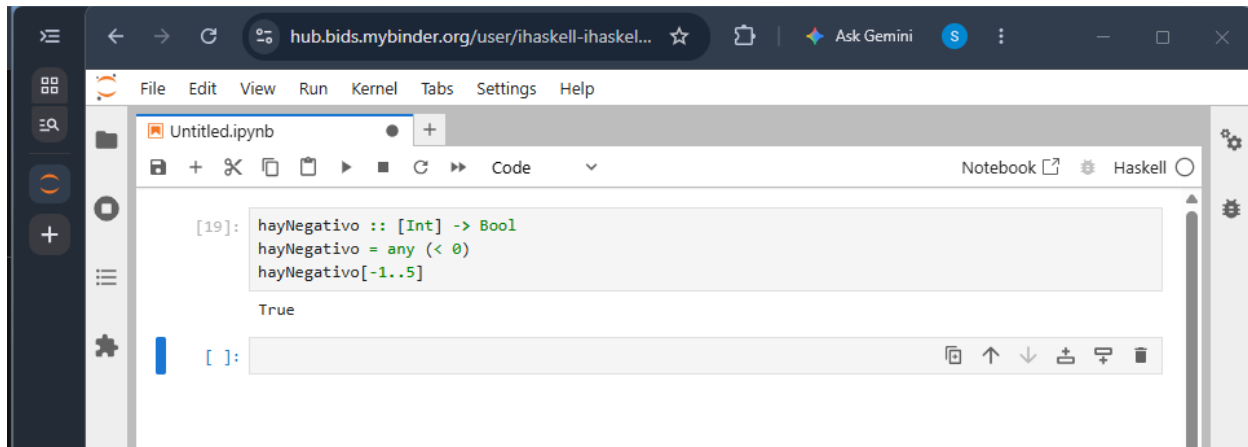
c) Dada una lista de números, devuelve los cuadrados de los pares.

The screenshot shows the same web browser window. The Jupyter Notebook is titled "Untitled.ipynb" and is in "Code" view. The code cell [16] contains the following Haskell code:

```
cuadradosPares :: [Int] -> [Int]
cuadradosPares = map (^2) . filter even
cuadradosPares [1,2,3,4,5]
```

The output of the code is `[4,16]`.

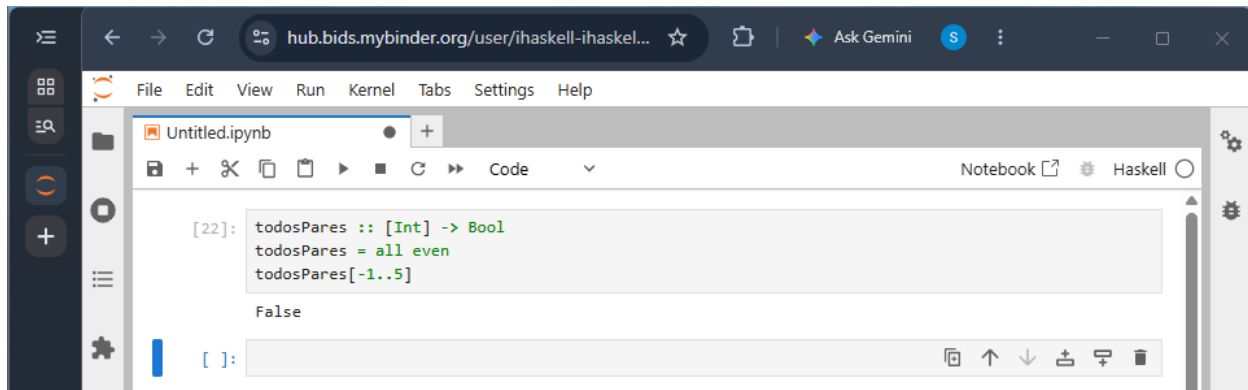
- 4) Define las siguientes funciones usando any o all
- a) Verifica si algún número en la lista es negativo.



```
[19]: hayNegativo :: [Int] -> Bool
      hayNegativo = any (< 0)
      hayNegativo [-1..5]

      True
```

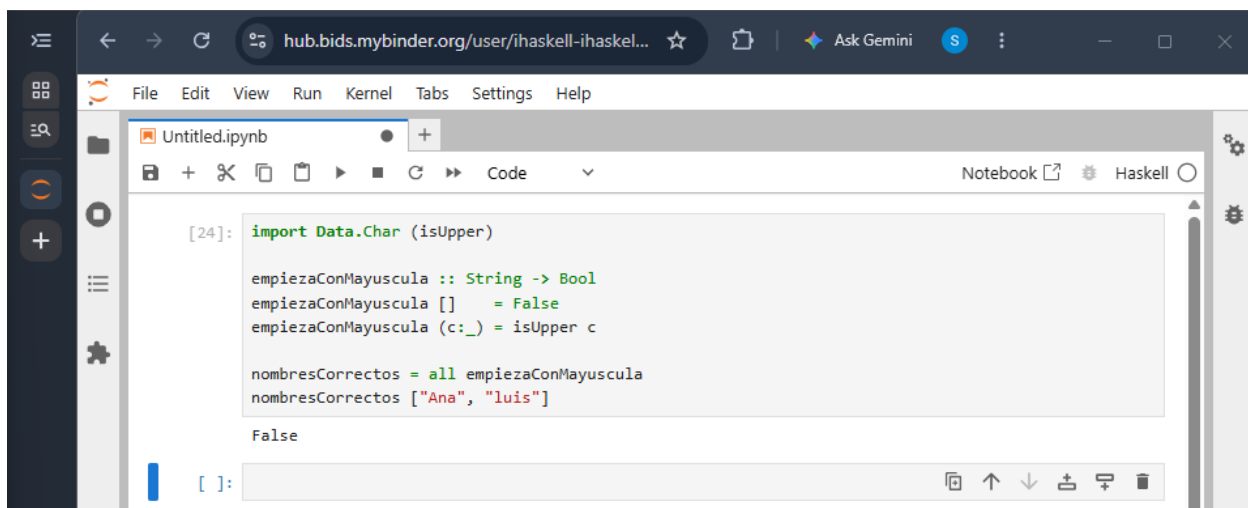
- b) Verifica si todos los números son pares.



```
[22]: todosPares :: [Int] -> Bool
      todosPares = all even
      todosPares [-1..5]

      False
```

- c) Comprueba si todos los nombres empiezan con mayúscula.



```
[24]: import Data.Char (isUpper)

      empiezaConMayuscula :: String -> Bool
      empiezaConMayuscula [] = False
      empiezaConMayuscula (c:_) = isUpper c

      nombresCorrectos = all empiezaConMayuscula
      nombresCorrectos ["Ana", "luis"]

      False
```

- 5) Define las siguientes funciones usando takeWhile o dropWhile
- a) De una lista ordenada, toma los elementos menores que 100.

The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The browser's address bar also contains "Ask Gemini" and a blue "S" icon. The Jupyter Notebook interface has a menu bar with "File", "Edit", "View", "Run", "Kernel", "Tabs", "Settings", and "Help". The notebook title is "Untitled.ipynb". The code cell [27] contains the following Haskell code:

```
menoresQue100 :: [Int] -> [Int]
menoresQue100 list = takeWhile (<100) list
menoresQue100 [50..100]
```

Below the code, the "Eta reduce" section shows the result of the function application:

```
Found:
menoresQue100 list = takeWhile (< 100) list
menoresQue100 = takeWhile (< 100)

[50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,
82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99]
```

The "Why Not:" section is empty. The bottom of the code cell shows a prompt `[ ]:` and a toolbar with icons for copy, paste, undo, redo, and delete.

b) Quita los ceros iniciales de una lista.

The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The browser's address bar also contains "Ask Gemini" and a blue "S" icon. The Jupyter Notebook interface has a menu bar with "File", "Edit", "View", "Run", "Kernel", "Tabs", "Settings", and "Help". The notebook title is "Untitled.ipynb". The code cell [29] contains the following Haskell code:

```
sinCerosIniciales :: [Int] -> [Int]
sinCerosIniciales list = dropWhile (==0) list
sinCerosIniciales [0,0,0,0,0,1,2,3,4]
```

Below the code, the "Eta reduce" section shows the result of the function application:

```
Found:
sinCerosIniciales list = dropWhile (== 0) list
sinCerosIniciales = dropWhile (== 0)

[1,2,3,4]
```

The "Why Not:" section is empty. The bottom of the code cell shows a prompt `[ ]:` and a toolbar with icons for copy, paste, undo, redo, and delete.

Ejercicio 4 - Funciones que reciben y retornan funciones

- 1) Define una función que recibe una función que transforma un dato (por ejemplo, conversión de grados Celsius a Fahrenheit), y lo aplica a una lista de datos.

The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The notebook interface has a menu bar with `File`, `Edit`, `View`, `Run`, `Kernel`, `Tabs`, `Settings`, and `Help`. The notebook title is `Untitled.ipynb`. The code cell contains the following Haskell code:

```
[32]: procesarLecturas :: (Double -> Double) -> [Double] -> [Double]
      procesarLecturas f lecturas = map f lecturas

      celsiusAFahrenheit :: Double -> Double
      celsiusAFahrenheit c = c * 9 / 5 + 32
      procesarLecturas celsiusAFahrenheit [0, 12, 20]
```

Below the code, the **Eta reduce** section shows the **Found:** result:

```
procesarLecturas f lecturas = map f lecturas
procesarLecturas celsiusAFahrenheit [0, 12, 20]
[32.0,53.6,68.0]
```

The **Why Not:** section is empty.

2) Define una función que retorne un filtro personalizado:

The screenshot shows the same web browser window. The notebook title is `Untitled.ipynb`. The code cell contains the following Haskell code:

```
[35]: crearFiltro :: (a -> Bool) -> ([a] -> [a])
      crearFiltro p = filter p

      soloImpares :: [Int] -> [Int]
      soloImpares = crearFiltro odd

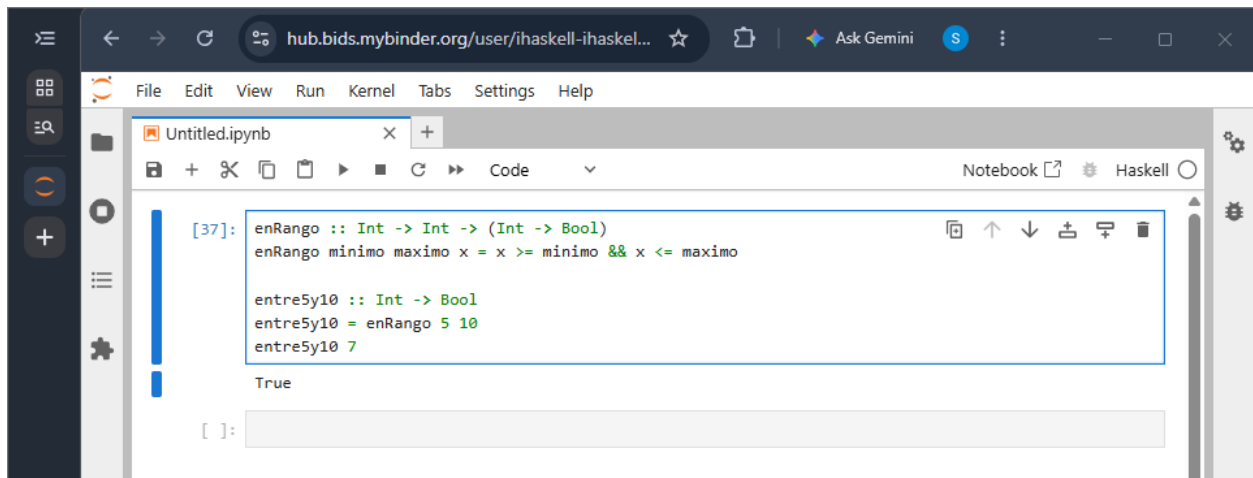
      soloImpares [1..10]
```

Below the code, the **Eta reduce** section shows the **Found:** result:

```
crearFiltro p = filter p
crearFiltro = filter
[1,3,5,7,9]
```

The **Why Not:** section is empty.

3) Crea una función que genera “verificadores” de rangos.



The screenshot shows a web browser window with the URL `hub.bids.mybinder.org/user/ihaskell-ihaskel...`. The browser's address bar also contains "Ask Gemini" and a blue circle with an "S". The Jupyter Notebook interface has a menu bar with "File", "Edit", "View", "Run", "Kernel", "Tabs", "Settings", and "Help". The notebook is titled "Untitled.ipynb" and is in "Code" mode. The code in the cell is as follows:

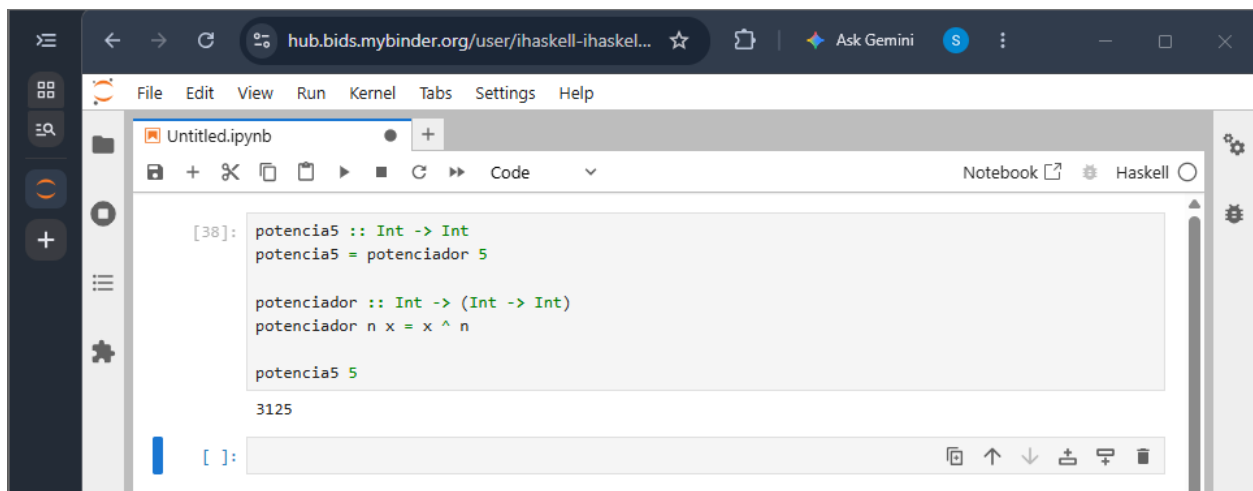
```
[37]: enRango :: Int -> Int -> (Int -> Bool)
      enRango minimo maximo x = x >= minimo && x <= maximo

      entre5y10 :: Int -> Bool
      entre5y10 = enRango 5 10
      entre5y10 7

      True
```

The output of the code is `[ ]:` followed by a text input field.

4) Genera una función que eleva a una potencia dada.



The screenshot shows the same Jupyter Notebook interface. The code in the cell is as follows:

```
[38]: potencia5 :: Int -> Int
      potencia5 = potenciador 5

      potenciador :: Int -> (Int -> Int)
      potenciador n x = x ^ n

      potencia5 5

      3125
```

The output of the code is `[ ]:` followed by a text input field.

Ejercicio 5 - “El problema de las dos jarras” usando A*

```
[43]: import Data.List (sortOn)

type Estado = (Int, Int)
type Nodo   = Estado
type Costo  = Int

capA = 5
capB = 3

transferirAB :: Estado -> Estado
transferirAB (a, b) = (a - q, b + q)
  where q = min a (capB - b)

transferirBA :: Estado -> Estado
transferirBA (a, b) = (a + q, b - q)
  where q = min b (capA - a)

vecinos :: Estado -> [(Estado, Costo)]
vecinos (a, b) =
  [ ((capA, b), 1)      -- Llenar A
  , ((a, capB), 1)      -- Llenar B
  , ((0, b), 1)         -- Vaciar A
  , ((a, 0), 1)         -- Vaciar B
  , (transferirAB (a, b), 1) -- Verter A->B
  , (transferirBA (a, b), 1) -- Verter B->A
  ]

heuristica :: Estado -> Costo
heuristica e
  | e == (4, 0) = 0
  | otherwise  = 1

aStarCamino :: (Nodo -> [(Nodo, Costo)]) -> (Nodo -> Costo) -> Nodo -> Nodo
              -> Maybe [(Nodo, Costo)]
aStarCamino sucesores h inicio meta = buscar [[(inicio), 0]] []
  where
    buscar [] _ = Nothing
    buscar ((camino@(nodo:_), costo) : cola) visitados
      | nodo == meta = Just (reverse camino, costo)
      | nodo `elem` visitados = buscar cola visitados
      | otherwise =
          let nuevos = [ (v : camino, costo + c)
                        | (v, c) <- sucesores nodo, v `notElem` visitados ]
              ordenados = sortOn \((p, g) -> g + h (head p)) (cola ++ nuevos)
          in buscar ordenados (nodo : visitados)

(transferirAB (5, 0), transferirBA (2, 3), transferirBA (0, 2), aStarCamino vecinos heuristica (0, 0) (4, 0))
((2,3),(5,0),(2,0),Just ([ (0,0), (5,0), (2,3), (2,0), (0,2), (5,2), (4,3), (4,0) ], 7))
```

Retrospectiva

- ¿Cuál fue el tiempo total invertido en el laboratorio por cada uno de ustedes? (Horas/Hombre)

Santiago Murillo 5 horas

- ¿Cuál es el estado actual del laboratorio? ¿Por qué?

Se encuentra completo, ya que cada punto se realizó de manera completa adjunta pruebas de que ejecuta y resuelve los enunciados

- ¿Cuál consideran fue el mayor logro? ¿Por qué?

Implementar una heurística con el algoritmo de Dijkstra, ya que, aunque esta búsqueda sea familiar todavía es relativamente nuevo mirar un nuevo método de solucionar problemas

- ¿Cuál consideran que fue el mayor problema técnico? ¿Qué hicieron para resolverlo?

Implementar el algoritmo del último punto, ya que este enfrenta una dificultad considerable a la hora de plasmar la idea en código

- ¿Qué hicieron bien como equipo? ¿Qué se comprometen a hacer para mejorar los resultados?
- ¿Qué referencias usaron? ¿Cuál fue la más útil? Incluyan citas con estándares adecuados.

<https://www.haskell.org/documentation/>

<https://claude.ai/artifact/NuVMB5iEVzwszbaNHT8GLM>

<https://claude.ai/artifact/1pGauKaw72zM5VQcsachj4>

Se realizó con ayuda de implementación de IA, (claude)